

Raj Kumar Mahto

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<https://rajmahto9876.github.io/Blog/resume/>

Professional Summary

Embedded Software Engineer with 5+ years of experience in firmware development, SoC bring-up, pre-silicon validation, and embedded systems debugging for automotive and IoT platforms. Strong expertise in Embedded C/C++, ARM Cortex-M/Cortex-A architectures, peripheral driver development, and low-level hardware interfacing including SPI, I2C, UART, DMA, timers, and interrupt subsystems. Experienced in RTOS-based systems, Linux kernel development, board bring-up, and register-level debugging using Lauterbach Trace32, JTAG, and GDB. Proficient in Git-based development workflows, embedded testing, and hardware schematic analysis.

Education

National Institute of Technology, Warangal	2025 – 2026
M.Tech(PG Diploma) in Control and Automation	CGPA: 8.37
Maharaja Agrasen Institute of Technology, Delhi	2016 – 2020
B.Tech in Electrical and Electronics Engineering	CGPA: 7.99

Technical Skills

Programming Languages: Embedded C, C++, Python
Embedded Systems: Bare-metal, Startup code, SoC Bring-up, Board-Bring Up
MCU / SoCs: ARM Cortex-M, ARM Cortex-A, NXP S32K5xx, ESP32, STM32, Atmel
Protocols & Networking: UART, SPI, I2C, DMA, MQTT, UDP, HTTPS, Wi-Fi (ESP32)
Operating Systems: FreeRTOS, Linux Kernel Modules, Character Device Drivers
Build & Toolchains: CMake, LLVM, Makefiles, Keil IDE, MPLABX
Debug & Validation Tools: Lauterbach Trace32, Veloce Emulation, Gdb, Valgrind
Version Control: Git and Bitbucket
Hardware tools: oscilloscope, Multimeter, Logic Analyzer, ST-Link, ULINK
Simulation tools: Matlab, Simulink
Code Quality & Compliance: MISRA-C coding guidelines, Static code analysis.

Professional Experience

VVDN Technologies (Onsite: NXP Semiconductors) May 2024 – July 2025
Senior Software Engineer

- Designed, developed, tested, and debugged embedded firmware for ARM-based automotive SoCs including peripheral validation, interrupt verification, timer subsystem testing, and low-level register debugging.
- Performed pre-silicon validation on the Veloce emulation platform, validated 8+ IP blocks some of them are interrupt controllers, timers, clock modules, and reset logic. Utilized Lauterbach Trace32 (T32) for register-level debugging, instruction tracing.
- Identified and resolved multiple logic and integration issues prior to silicon tape-out, improving hardware-software stability. Automated testing reduced regression time by 20%

IoTfy Solutions Pvt. Ltd., Delhi May 2021 – May 2024
Tech Lead – Embedded Systems

- Worked closely with hardware teams to analyze schematics, controller datasheets, and peripheral specifications during board bring-up and product development.

- Mentored and guided a team of embedded engineers, enforcing coding standards, design reviews, and best practices.
- Designed **FreeRTOS-based task** architecture using queues, semaphores, mutexes, and software timers.

Copper Connection Pvt. Ltd., Delhi

April 2020 – May 2021

Embedded Engineer

- Worked on embedded firmware development and feature implementation for PBX systems.
- Supported hardware prototyping, debugging, and system validation activities.

Key Projects

Hardware Security Engine (HSE) – Pre-Silicon Project

May 2024 – June 2025

- Developed and maintained **CMake build scripts** for compiling bare-metal firmware in a multi-module environment.
- Worked extensively on **linker scripts and startup files** for system initialization and memory mapping. Enabled and validated critical **bare-metal security and system IPs** including Interrupt, Timer, Clock Subsystems, MC_RGM, and MC_ME.
- Implemented and executed **unit and system-level test cases** to verify firmware correctness and IP behavior. Performed **system testing in a pre-silicon environment** using the Veloce emulation platform.

IoT Switches, Plugs, Wi-Fi Mesh Network Implementation (ESP32 & Chimney Systems)

May 2023 – May 2024

- Designed and implemented a Wi-Fi Mesh network supporting 40+ devices under a single IP.
- Designed and developed an **IoT-enabled smart devices** with mobile app, HTTP, web server, UDP, MQTT, and cloud integration.
- Implemented **JSON-based** data formatting for efficient and reliable device-to-cloud communication.
- Enabled real-time monitoring and remote control through cloud-connected embedded firmware.

BLDC Fan Development

May 2021 – May 2024

- Optimized BLDC fan tuning and wattage control algorithms to achieve a **5-star efficiency rating**.
- Assisted in hardware design activities including **PCB layout** and **SMPS integration**. Implemented **low-voltage and high-voltage cut-off protection circuits** to enhance system safety and reliability.
- Developed **multi-mode control functionality**, supporting IR remote, RF remote, and cloud-based operation.

Positions of Responsibility

- Technical Lead, IoTfy Solutions – 3 Years
- Senior Embedded Developer, NXP (via VVDN) – 1 Year

Languages

- English – Professional Working Proficiency
- German – Basic A1 Proficiency